

FoodPro Database Tables, Workflow, and Schema

Source: generated from the current FoodPro implementation in server/sql/init.sql, server/entities.js, server/index.js, server/locationScope.js, and workflow screens in Production, Material Requests, and Procurement.

1. SQL Tables

- users: Application users, roles, site assignment, password hash, profile metadata
- auth_tokens: Login session tokens
- entity_records: Hybrid JSONB-backed business entities
- app_logs: Application activity log
- email_logs: Email dispatch log
- pos_sources: POS providers and sync settings
- pos_sales_orders: Imported POS order headers
- pos_sales_items: Imported POS order lines
- pos_recipe_mapping: POS item to recipe mapping
- pos_sync_logs: POS sync execution log
- suppliers: Supplier master
- purchase_requests: Procurement request header
- purchase_request_items: Procurement request lines
- purchase_orders: Purchase order header
- purchase_order_items: Purchase order lines
- goods_receipts: GRN header
- goods_receipt_items: GRN lines with batch and expiry
- supplier_invoices: Supplier invoice header
- supplier_price_history: Historic supplier ingredient prices

2. Entity-Backed Schema Models

- Site: Project and hierarchy master with project_code, parent, company/region/location/kitchen/store structure
- User: Operational user profile, allowed projects, visibility scope, role permissions
- RoleProfile: Custom role and permission profile
- Ingredient: Ingredient master, category, cost, nutrition, allergens, shrinkage

- Recipe: Recipe master, servings, ingredients, nutrition, site scope
- Inventory: Current stock by site and ingredient
- InventoryLot: Batch/lot stock with expiry
- InventoryTransaction: Ledger movement for receipts, issues, adjustments, transfers, production
- MenuPlan: Planned menu by site/date/meal type
- Production: Production request/batch execution record
- MaterialRequest: Linked material request for shortages / procurement handoff
- FoodWaste: Waste posting by ingredient/recipe/batch/location with approval and cost
- WasteTarget: Waste reduction targets by site and month
- ProductionTransfer: Finished goods transfer between locations
- Supplier: Entity-backed supplier records used in some modules
- PurchaseOrder: Entity-backed PO records used by app workflows
- QualityControl: Quality checks and compliance records
- AttendanceSession: Attendance / QR operational session
- AttendanceRecord: Staff attendance detail
- StaffShift: Scheduling and shift plan
- ForecastScenario: Demand forecast scenario definition
- ForecastSnapshot: Generated forecast result snapshot
- AdvancedReportSchedule: Scheduled report email definition
- ERPIntegrationConfig: ERP provider endpoint and mapping settings
- ERPIntegrationLog: ERP export log and retry tracking
- CustomerMealPlan: Nutrition and meal planning records
- BranchOrder: Branch demand/order capture
- QRCode: QR definition records
- QRDelivery: QR delivery records
- RFQ: Request for quotation record

3. Architecture and Relationship Notes

- The platform uses a hybrid schema: PostgreSQL relational tables for auth, POS, and procurement; JSONB entity_records for flexible business modules.
- Inventory uniqueness is enforced at business level by site_id + ingredient_id. Inventory lots extend stock with batch_number and expiry_date.

- Production links to recipe, site, meal_type, target_servings, ingredients_used, costing, and linked material request lifecycle.
- FoodWaste can link to ingredient, recipe, production batch, or location-level waste and stores estimated_cost, approval_status, avoidable_type, and batch_reference.
- User visibility is location-scoped using allowed_site_ids and visibility_scope; server-side filtering is applied through locationScope.js.
- RoleProfile and system role definitions combine for effective permissions such as create_production_request, approve_waste, acknowledge_material_request, manage_procurement, and manage_erp.

4. Production Approval + Material Request Workflow

- 1. Chef creates production request with recipe, site/project, meal type, servings, and ingredient requirement snapshot.
- 2. System generates a linked draft material request from production ingredients.
- 3. Chef submits production request for approval.
- 4. Project Manager / Operations Manager reviews the request and can approve, reject, or request changes.
- 5. When approved, the linked material request is released to procurement with status pending_procurement_ack.
- 6. Procurement Officer acknowledges the material request.
- 7. After acknowledgement, chef can start production.
- 8. On completion, inventory is consumed and production cost is posted.

5. Procurement Workflow

- 1. Purchase Request is created manually or auto-generated from low stock.
- 2. Purchase Request is reviewed and approved according to role permissions.
- 3. Approved request is converted to Purchase Order.
- 4. Goods Receipt Note is posted on receipt with accepted/rejected quantity, batch number, and expiry date.
- 5. Inventory is updated from GRN into stock and inventory lots.
- 6. Supplier Invoice is entered and matched to PO / GRN.
- 7. Supplier price history is updated for future variance reporting.

6. Food Waste Workflow

- 1. Waste is posted by ingredient, recipe, batch, or location.
- 2. System calculates estimated waste cost from ingredient cost or recipe / batch cost logic.

- 3. If waste value exceeds approval threshold, approval_status becomes pending.
- 4. Authorized manager/quality approver can approve or reject high-value waste.
- 5. Waste data feeds trends, reason analysis, avoidable vs unavoidable analysis, and waste reduction intelligence.
- 6. Production planning can display historical waste intelligence by recipe and meal type.

7. Inventory Workflow

- 1. Inventory enters from GRN/receipts, manual stock addition, or seeded stock.
- 2. Lots carry batch_number, expiry_date, valuation, and remaining quantity.
- 3. Stock moves through transfers, adjustments, issues, production consumption, and POS deductions.
- 4. Reports include stock on hand, movement ledger, expiry, valuation, and velocity.

8. Location + Permission Workflow

- 1. Hierarchy supports Company -> Region -> Location -> Kitchen -> Store/Warehouse.
- 2. Users are assigned to one or more projects/locations through allowed_site_ids.
- 3. Server filters records by accessible site scope for non-admin users.
- 4. Roles can be system-defined or custom through RoleProfile with granular permissions.

Summary

FoodPro uses a hybrid database strategy to support rapid module growth while preserving strong workflow control. Auth, POS, and procurement use dedicated relational tables; the main operational modules use entity_records with server-side validation, uniqueness rules, permission enforcement, and location scoping.